PHYSICS – Code No. 042

SAMPLE QUESTION PAPER

CLASS – XII (2025 – 26)

Time Allowed: 3 hours

Maximum Marks: 70

General Instructions

- There are 33 questions in all. All questions are compulsory.
- This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- All the sections are compulsory.
- Section A contains sixteen questions, twelve MCQ and four assertion reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study-based questions of four marks each and Section E contains three long answer questions of five marks each.
- There is no overall choice. However, an internal choice has been provided in two question in Section B, one question in Section C and all three questions in Section E. You have to attempt only one of the choices in such questions.
- Use of calculators is not allowed.
- You may use the following values of physical constants where ever necessary

Section - MCQ

1. Which of the following values of the refractive index n of a medium yields a total internal reflection at an incident angle of 30° from air (n=1)?

(A) 1.0
(B) 1.33
(C) 2.0
(D) 0.75

[1 marks]

2. In an interferometer using a coherent source of wavelength 600 nm, what is the path difference corresponding to the 5th bright fringe when the beam is reflected from a glass plate of thickness 0.5 mm and refractive index 1.5?

(A) 4.5 µm
(B) 3.0 µm
(C) 6.0 µm
(D) 1.5 µm

[1 marks]

3. Which of the following pairs of polarizers will produce the maximum transmitted intensity when the incident light is linearly polarized at 30° to the first polarizer's axis?

(A) Axes at 30° and 30°
(B) Axes at 30° and 60°
(C) Axes at 0° and 30°
(D) Axes at 0° and 90°

[1 marks]

4. In a circuit comprising a voltage source of 12 V, a resistor R1=4 kΩ, and a resistor R2 in series, the power dissipated in R2 is 24 W. What is the resistance value of R2?

(A) 1 kΩ
(B) 2 kΩ
(C) 3 kΩ
(D) 4 kΩ

[1 marks]

5. A long straight wire of radius 2 mm carries a current of 5 A. What is the magnitude of the magnetic field 5 mm away from the centre of the wire?

(A) 3.95 x10^-6 T
(B) 4.42 x10^-6 T
(C) 1.57 x10^-6 T
(D) 2.09 x10^-6 T

[1 marks]

6. A loop of area 0.01 m^2 is placed in a magnetic field of 0.2 T. If the loop rotates at 10 rad/s, what is the maximum emf induced?

(A) 4 V
(B) 2 V
(C) 1 V
(D) 0.5 V

[1 marks]

7. Which of the following combinations of conductors and insulators will give the highest resistance between two points separated by 10 cm at 25°C?

(A) Copper wire 2 mm^2
(B) Aluminium wire 2 mm^2
(C) Graphite plate 2 mm^2
(D) Rubber strip 2 mm^2

[1 marks]

8. A magnetic field inside a toroid with N=200 turns and radius 10 cm is 0.8 T. What is the current passing through the toroid?

(A) 1 A
(B) 2 A
(C) 4 A
(D) 8 A

[1 marks]

9. A monochromatic light of wavelength 500 nm undergoes diffraction by a single slit of width 0.1 mm. What is the angular position of the first minimum?

(A) 5.7°
(B) 3.6°
(C) 1.4°
(D) 0.5°

[1 marks]

10. What is the value of the electric field at a point 5 cm from a uniformly charged rod of length 10 cm and linear charge density 2 μC/m?

(A) 1.8 kV/m
(B) 3.6 kV/m
(C) 2.4 kV/m
(D) 4.5 kV/m

[1 marks]

11. Which of the following statements about the skin effect in conductors at high frequency is FALSE?

(A) The current density is higher near the surface.
(B) The effective resistance increases with frequency.
(C) The penetration depth decreases with frequency.
(D) The skin depth is independent of material resistivity.

[1 marks]

12. A capacitor of capacitance 100 μF is connected to a 12 V battery. What is the energy stored?

(A) 0.72 J
(B) 1.44 J
(C) 7.2 J
(D) 14.4 J

[1 marks]

13. Which of the following wave phenomena cannot be explained by the Huygens principle?

(A) Diffraction through a slit
(B) Reflection from a flat surface
(C) Refraction at an interface
(D) Polarization by a polarizer

[1 marks]

14. A 5 m long solenoid has 800 turns and carries a current of 2 A. What is the magnetic field inside the solenoid?

(A) 2.0 T
(B) 0.4 T
(C) 1.6 T
(D) 3.2 T

[1 marks]

15. If the potential difference across a 3 Ω resistor is 9 V, what is the current through it?

(A) 0.3 A
(B) 1 A
(C) 3 A
(D) 9 A

[1 marks]

16. Which of the following is the correct expression for the magnetic force on a straight conductor of length L carrying current I in a magnetic field B perpendicular to the conductor?

(A) F = I L B
(B) F = B I / L
(C) F = I B L / 2
(D) F = I B / L

[1 marks]

17. A light wave of wavelength 600 nm undergoes total internal reflection inside a prism. What is the minimum angle of incidence required for total internal reflection?

(A) 30°

(B) 45°

(C) 60°

(D) 75°

[1 marks]

18. The induced emf in a coil of 200 turns when the magnetic field changes from 0.5 T to 0.2 T in 0.05 s is?

(A) 1200 V

(B) 800 V

(C) 600 V

(D) 400 V

[1 marks]

19. Which of the following materials would be the best choice for an electrical insulator at 300 K?

(A) Glass

(B) Copper

(C) Aluminium

(D) Carbon

[1 marks]

20. A magnetic dipole moment $\mu = 3.2 \times 10^{-5}$ A·m^2 is placed in a uniform magnetic field of 0.01 T. What is the torque on it when the dipole is at 45° to the field?

(A) 2.26 x10^-7 N·m

(B) 4.53 x10^-7 N·m

(C) 1.07 x10^-6 N·m

(D) 8.53 x10^-7 N·m

[1 marks]

Section - Short Answer

21. Explain the phenomenon of total internal reflection and its application in optical fibers.

[3 marks]

22. Derive the expression for the resistivity of a conductor from Ohm's law and explain how temperature affects it.

[3 marks]

23. Describe how a magnetic field exerts a force on a moving charge and provide the equation for the magnitude of this force.

[3 marks]

24. Explain the principle behind the operation of an induction cooker and identify the role of the alternating current.

[3 marks]

25. Calculate the wavelength of light emitted by an electron transitioning from n=3 to n=1 in a hydrogen atom using the Bohr model.

[3 marks]

26. Discuss the concept of magnetic flux and how it relates to Faraday's law of induction.

[3 marks]

27. Define the skin effect in conductors and state how it influences the AC resistance of a wire.

[3 marks]

28. Explain how a photomultiplier tube amplifies a single photon signal and list its main components.

[3 marks]

29. Derive the expression for the magnetic field inside an infinite solenoid and explain why it is uniform.

[3 marks]

30. Illustrate the principle of operation of a Hall effect sensor and identify how the Hall voltage depends on current and magnetic field.

[3 marks]

Section - Long Answer

31. Derive the condition for constructive interference in a double-slit experiment when one of the slits is replaced by a grating of N equally spaced slits, and explain how the interference pattern changes as N increases.

[5 marks]

32. Explain how the magnetic field inside a toroid differs from that outside, and derive the expression for the magnetic field inside using Ampere's law.

[5 marks]

33. Derive the formula for the impedance of a series RL circuit and discuss the effect of frequency on its magnitude.

[5 marks]

34. Describe how the polarization of light is altered when it passes through a birefringent crystal oriented at 45° to the incident polarization, and calculate the phase difference between ordinary and extraordinary rays.

[5 marks]

35. Using Kirchhoff's laws, find the equivalent resistance of a complex network consisting of a resistor of 10 Ω in parallel with a series combination of a 5 Ω resistor and a 15 Ω resistor.

[5 marks]

36. Explain the principle of operation of a Hall effect sensor and derive the expression for Hall voltage in terms of current, magnetic field, carrier density, and thickness of the material.

[5 marks]